

InventOR: A Human-Reviewed, Prompt-Configured Large Language Model Decision Support System for Material-Level Inventory Control

Aris Dressino^{a,*}

^a*TUM School of Management, Technical University of Munich, Arcisstraße
21, Munich, 80333, Germany*

Abstract

Material planners in manufacturing must set a reorder point, a safety stock and an order quantity for thousands of plant-material pairs, usually under ERP parameters that were last revised long ago. We define, build and evaluate *InventOR*, a decision support system (DSS) for this recurring, semi-structured operational decision. The planner remains the decision-maker. For each material the system supplies a candidate reorder-point/order-quantity (r, Q) policy generated by a large language model (LLM) that is configured only through a prompt, without training or fine-tuning, together with a written justification, an escalation flag and a counterfactual simulation. Deterministic software parses every proposal, gates it against minimum order quantities and storage limits, and replays it on held-out post-cutoff demand before the planner sees it. We derive six design requirements and evaluate the artifact on 365 plant-material pairs from three plants of an industrial manufacturer (346 in the common-feasible cohort), using three independent LLM runs, seven reference arms that

*Corresponding author. Telephone: +49 89 289 25000. ORCID: 0009-0003-0693-9051.
Email address: aris.dressino@tum.de (Aris Dressino)

differ in the information and the method they use, and an ablation that withholds the ERP planning parameters. Against ERP-derived parameters the LLM policies cost about 11% less in simulation at a service level 0.1–0.4 percentage points lower. Data-driven operations-research (OR) baselines computed from the history alone, however, cost less than the LLM at the same service: about 6% less for a normal-approximation policy, about 8% less for a Croston/SBA policy and about 18% less for an empirical lead-time-demand quantile policy. Much of the apparent gain therefore reflects weak ERP benchmarks rather than optimization by the LLM; the saving over ERP parameters comes from holding about 17% less inventory at about 13% more stockout days, and when shortages are priced at one unit cost or more the ERP-derived parameters are the cheapest of the non-tuned arms while a simulation-tuned policy is cheapest overall. Withholding the ERP parameters leaves the LLM about 9% cheaper than ERP-derived parameters and about 9–21% more expensive than the history-only OR baselines. Storage limits that appear satisfied in a static check are exceeded on 53–66% of pairs along unconstrained simulated trajectories, so a hard-limit replay is our primary evaluation. At one plant the LLM is 31–36% more expensive than ERP-derived parameters, a result that is robust to our calendar repairs. The evidence supports InventOR as a human-reviewed DSS whose value lies in producing an auditable candidate policy for every material, not in out-optimizing a competent data-driven OR baseline. We contribute design principles for LLM-based DSS, explicit trust, escalation and rejection rules, and a research agenda.

Keywords: Decision Support Systems, Large Language Models, Generative Artificial Intelligence, Inventory Control, Human–AI Decision Making, Design Science, Operations Research

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point that triggers an order, a safety stock that absorbs demand and lead-time variability, and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand patterns, supplier performance and storage constraints change. Revisiting them materially requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. That scarce skill is exactly what a conversational large language model (LLM) with a code interpreter appears to offer, and a rapidly growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5].

The promise is real, but so are three problems that limit its use in practice. First, LLM outputs can be unfaithful or fabricated [6], so a proposed policy must be treated as a candidate and not a decision. Second, claims of superiority over ERP parameters are hard to interpret when the LLM and the comparators are given different information; an LLM that sees a plant’s ERP reference values and is compared with an OR policy that does not, or the reverse, tells us little about the LLM. Third, inventory policies live inside physical limits: a storage rule that holds when the policy is written down can be violated as stock builds up over time.

This paper treats the problem as one of decision support system (DSS) design. We define *InventOR* as a human-reviewed, model-driven DSS for the recurring decision of setting replenishment parameters per plant-material pair (Section 2). A planner remains the decision-maker. The system supplies,

26 for each material, a candidate (r, Q) policy proposed by an enterprise LLM
 27 application that is configured only through a prompt and receives no project-
 28 specific training or fine-tuning, together with a justification, an escalation
 29 flag and a counterfactual simulation; deterministic software parses, gates
 30 and simulates every proposal before the planner reviews it and alone decides
 31 whether any ERP parameter is changed. The pattern is not specific to
 32 inventory: any production-planning decision in which parameters are set per
 33 item and refreshed periodically (lot sizes, planned lead times, buffer levels,
 34 dispatching rules) has the same structure of many semi-structured decisions,
 35 scarce analyst time, hard constraints and a need for accountable evidence.
 36 The design lessons below therefore concern LLM-based decision support for
 37 production and logistics planning in general, and not only inventory control.

38 We follow a design-science logic [7, 8]: we state the problem and its scope,
 39 derive design requirements, build the artifact, and evaluate it against those
 40 requirements on real industrial data. Four research questions structure the
 41 paper.

42 **RQ1. *Design.*** How should an LLM be embedded in a DSS so that its
 43 proposals are bounded, verifiable, reproducible and reviewable by a
 44 planner?

45 **RQ2. *Efficacy.*** What cost and service do the proposed policies achieve
 46 relative to ERP-derived parameters and to OR policies, when the
 47 information available to each approach is controlled?

48 **RQ3. *Feasibility and reliability.*** Do the policies respect physical storage
 49 limits along simulated inventory trajectories, and how stable are they
 50 across independent runs?

51 **RQ4.** *Use.* Under which conditions should a planner trust, escalate or reject
52 a proposal?

53 Our contributions are:

54 **C1. A definition and scope of the DSS** (Section 2): the decision,
55 the decision-maker, the system boundary, the level of automation and
56 six design requirements that make the role of the LLM explicit and
57 bounded.

58 **C2. A DSS architecture and operating rules** (Section 4) that separate
59 generation, deterministic gating, simulation and planner review, with
60 explicit rules for when output is trusted, escalated or rejected, and
61 design principles that generalize beyond this application (Section 8).

62 **C3. An information-controlled, trajectory-level evaluation** (Sec-
63 tion 5) with seven reference arms that differ in the information they use
64 and in method (ERP-derived, ERP-informed, normal-approximation,
65 Croston/SBA, empirical-quantile and simulation-tuned policies), an
66 LLM ablation denied the ERP planning parameters, and a hard-limit
67 storage replay that is the primary result. This separates what the
68 LLM adds from what it is given.

69 **C4. Complete-cohort industrial evidence** from three LLM runs on 365
70 plant-material pairs, with plant-level, regime-level and sensitivity anal-
71 yses, and managerial and research implications.

72 Our main findings are as follows. Against ERP-derived parameters, the
73 LLM policies are about 11% cheaper at a service level that is 0.1–0.4 points
74 lower. That advantage largely disappears when a reasonable OR baseline

75 is available: history-only (r, Q) baselines are cheaper than the LLM at es-
 76 sentially equal service (about 6% for a normal-approximation policy, 8% for
 77 Croston/SBA and 18% for an empirical lead-time-demand quantile policy),
 78 and the LLM’s saving over ERP parameters comes from carrying about 17%
 79 less inventory at about 13% more stockout days; once shortages are priced
 80 at one unit cost or more, the ERP-derived parameters are the cheapest of
 81 the non-tuned arms and a simulation-tuned policy is cheapest overall. The
 82 LLM’s value is therefore not superior optimization but the ability to produce
 83 a reviewable, auditable policy for every material without writing code. Stor-
 84 age limits are violated along unconstrained simulated trajectories for 53–66%
 85 of pairs under every method, which makes trajectory checks a prerequisite
 86 for any replenishment decision support system. At one plant the LLM is 31–
 87 36% more expensive than ERP-derived parameters, which we use to derive
 88 guidance on where to trust and where to review.

89 The remainder of the paper is organized as follows. Section 2 defines
 90 the problem, the scope of the DSS and its design requirements. Section 3
 91 positions the work in the literature. Section 4 describes the architecture and
 92 operation of the DSS for a non-specialist reader. Section 5 describes the
 93 data and the evaluation design. Section 6 reports results. Section 7 derives
 94 managerial insights, and Section 8 discusses design principles, limitations
 95 and research directions. Formal definitions are in Appendix Appendix A.

96 **2. Problem, Scope and Definition**

97 *2.1. The decision problem*

98 The decision supported is the periodic (re)setting of three replenishment
 99 parameters, a reorder point, a safety stock and an order quantity, for each

100 plant-material pair held in an ERP system. In the classification of Gorry
 101 and Scott Morton [9] it is an operational-control decision that is partly struc-
 102 tured: the calculation is textbook, but the choices of estimator, the treat-
 103 ment of intermittent or trending demand, the plausibility of recorded lead
 104 times and the handling of constraints require judgement that is applied in-
 105 consistently when thousands of materials compete for a few analysts' time.
 106 The decision-maker is the material planner or inventory analyst; the deci-
 107 sion is repeated, high-volume and consequential for working capital, service
 108 and storage. Hard constraints are the minimum order quantity (MOQ) and
 109 a physical storage limit. The feedback loop is slow: a changed parameter
 110 shows its effect only in subsequent demand.

111 2.2. Definition and level of automation

112 **Definition.** *InventOR is a human-reviewed, model-driven deci-*
 113 *sion support system that, for one plant-material pair at a time, (i)*
 114 *obtains from a large language model, configured only by a written*
 115 *prompt, a candidate (r, Q) policy with a justification and an es-*
 116 *calation flag; (ii) verifies the candidate deterministically against*
 117 *explicit constraints; (iii) evaluates it by counterfactual simulation*
 118 *against reference policies on held-out demand; and (iv) presents*
 119 *the result to a planner, who alone decides whether ERP parame-*
 120 *ters are changed.*

121 In the classical DSS architecture of data, model and dialog components
 122 [10, 11], the LLM and the simulator form the model component, the material-
 123 scoped data extract and the prompt form the data and knowledge compo-
 124 nents, and the review interface is the dialog component (Section 4.1). In

terms of the levels of automation of Parasuraman et al. [12], InventOR operates at a low-to-intermediate level: it selects and suggests one alternative and leaves execution to the human. This is a design decision, not a limitation: acceptance of automated suggestions is accompanied by complacency and automation bias [13], which the review rules of Section 4.3 are meant to counter.

2.3. Scope

Table 1 states what is and is not part of the system and of the evaluation, so that the claims of the paper are read at the right level.

Table 1: Scope of InventOR and of its evaluation.

Dimension	In scope	Out of scope
Decision	Reorder point, safety stock and order quantity per plant-material pair, reviewed periodically	Order release, supplier selection, pricing, forecasting-model development
Inventory system	Single item, single echelon, continuous review, lost sales, (r, Q) family	Multi-item coupling, shared capacity, multi-echelon networks
Inputs	Daily demand and realized lead times, working-day flags, cost and constraint parameters, ERP reference parameters	External signals, supplier forecasts, market data
Role of the LLM	Candidate generator with justification and escalation flag	Forecasting, order placement, running the simulation, final decision
Human role	Planner reviews and decides; rules for trust, escalation and rejection	Autonomous execution without review
Evaluation	Ex-post simulation on real post-cutoff demand, information-controlled comparison, hard storage-limit replay, repeated runs	Field trial, planner workload or decision quality, user acceptance

2.4. Design requirements

Four sources motivate the requirements: the known unreliability of LLM outputs [6], guidance on human–AI interaction and accountability [14, 15], the run-to-run nondeterminism of LLM inference [16], and the practical con-

138 constraints of inventory control. They yield six requirements (Table 2), each
 139 tied to a design feature (Section 4) and an evaluation (Section 6).

Table 2: Design requirements, the design features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.8
DR4	Information transparency: comparisons control what each arm sees	Four reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.7
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

140 2.5. Research approach

141 We follow the design-science research process of problem identification,
 142 requirements, artifact design and evaluation [7, 8]. In the evaluation frame-
 143 work of Venable et al. [17], our evaluation is artificial and ex post: it uses
 144 real industrial data and a faithful simulation but no live users. It therefore
 145 establishes technical efficacy, feasibility and reliability; it does not establish
 146 usability, adoption or improved planner decisions, which we list as limitations
 147 and research directions.

148 3. Related Work and Positioning

149 *Decision support systems and human-AI decision making.* DSS research
 150 has long distinguished data-, model- and knowledge-driven systems and has
 151 stressed that the value of a DSS lies in improving the judgement of a human
 152 decision-maker rather than in replacing it [9, 10, 11]. LLMs add a new kind

153 of model component: it is configured through language, produces free-text
154 justification and is not deterministic. Research on human–automation in-
155 teraction shows that the way suggestions are scoped, presented and verified
156 determines whether users rely on them appropriately or over-rely on them
157 [12, 13], and guidelines for human–AI interaction call for explicit scoping,
158 explanation and correction mechanisms [14]. Our contribution to this stream
159 is a concrete, evaluated instance of an LLM-based DSS for an operational
160 production decision, with design requirements that are tied to evidence.

161 *LLMs in operations and supply-chain decisions..* Reviews and frameworks
162 map the opportunities for generative AI in supply chains [1, 2, 3, 4, 18]
163 and stress that adoption depends on assigned human roles, data quality and
164 governance [5, 19, 20, 21]. A recent conceptual line proposes agentic digital
165 twins that combine AI agents with model-based simulation and optimiza-
166 tion to support decisions [22], and an LLM-plus-digital-twin method has
167 been built for closed-loop supply-chain operations [23]. Our decision sup-
168 port system follows the same logic at a smaller scale: an LLM proposes and
169 a simulation model checks. Most technical LLM work focuses on translating
170 natural-language problem statements into optimization models and solver
171 code [24, 25, 26, 27], on multi-agent control and consensus-seeking in inven-
172 tory flows [28, 29], on extracting supply-network structure from text [30, 31],
173 on disruption modelling [32], or on evaluating LLMs on classical operations-
174 research tasks such as vehicle routing [33]. The closest LLM study to ours
175 uses an LLM as a forecasting copilot for spare-parts demand, combined with
176 simulation, and reports high variability for irregular demand that can be
177 reduced by averaging repeated queries and enriching prompts [34]. We share
178 the concern about run-to-run variability and test it directly (Section 6.7),

179 but differ in that the LLM proposes a complete replenishment policy rather
180 than a forecast.

181 *Data-driven inventory control without LLMs..* Learning-based replenish-
182 ment is well established. Deep reinforcement learning has been applied to
183 omnichannel and multi-echelon systems [35, 36] and aerospace replenishment
184 [37]; deep-learning forecasting has been integrated with replenishment [38];
185 contextual distributionally robust optimization uses side information to
186 set order quantities [39]; and distribution-free online algorithms handle
187 capacitated warehouses under uncertain price and demand [40]. Robust
188 inventory control under demand and lead-time uncertainty has a longer
189 history [41, 42, 43], and lead-time variability and dependence are known
190 to propagate through ordering decisions [44], while safety stock and safety
191 lead time are alternative buffers against unreliability [45]. The closest
192 operational neighbour is a trained data-driven framework that jointly
193 sets replenishment parameters, safety stock and safety lead time on real
194 automotive data [46]. These methods require model training, historical
195 scenario generation or tuned hyperparameters for each setting; InventOR
196 requires none, at the price of an LLM-based generator whose internal logic
197 cannot be inspected beyond its written justification.

198 *Intermittent and spare-parts demand..* Material histories in manufacturing
199 are frequently intermittent or lumpy. The statistical literature offers special-
200 ized estimators and classification schemes [47, 48, 49, 50, 51], and recent stud-
201 ies develop feature-based forecast combinations and temporal aggregation for
202 intermittent demand [52, 53], forecasting for spare parts across product life-
203 cycle phases [54], and integrated forecasting and planning or multi-echelon
204 stocking models for repairable and spare parts [55, 56]. Our regime analysis

205 (Section 6.6) shows where an LLM that is forbidden from using specialized
206 intermittent-demand estimators behaves differently on lumpy materials.

207 Table 3 summarizes these streams and states how the contribution of
208 InventOR differs. InventOR is not a new inventory model and does not
209 claim to beat trained or optimized methods. It contributes (i) a DSS design
210 in which an off-the-shelf LLM is a bounded candidate generator inside a
211 reviewed process, (ii) an evaluation that controls the information available to
212 each arm, and (iii) industrial evidence on cost, service and storage feasibility.
213 The (r, Q) policy it uses is the standard workhorse of continuous-review
214 control [57, 58, 59].

Table 3: Literature positioning of InventOR relative to recent work on AI-supported inventory and supply-chain decisions.

Study	Approach	Decision	Relation to InventOR
<i>LLMs for operations and supply chains</i>			
Jackson et al. [1]; Fosso Wamba et al. [2]	Framework / review	Broad supply-chain use	Motivates LLM use; no inventory-policy evidence
Song et al. [3]; Panja et al. [4]	Systematic reviews	LLMs in SCM and logistics	Place InventOR in the emerging empirical gap
Trifone et al. [5]; Cannas et al. [19]; Venkatesh et al. [20]	Adoption studies, search agenda	AI roles, barriers, data quality	Support human-in-the-loop design and review rules
Ivanov [22]; Le et al. [23]	Agentic LLM digital twins	Model-based decision support	Pair AI proposals with simulation; InventOR applies this to (r, Q) parameters
Nakhost et al. [34]	LLM + simulation	Spare-parts forecast	Closest LLM study; forecast, not policy
Jannelli et al. [29]; Quan and Liu [28]	Multi-agent LLM	Sequential ordering, consensus	Per-period agent decisions in simulation
AlMahri et al. [30]; Zheng and Brintrup [31]	LLM + knowledge graphs	Supply-network visibility	LLM for information extraction, not inventory control
Hamdan and Brahimi [33]	LLM benchmark	Routing	Evaluates LLM on a classical OR task
Li et al. [24]; Huang et al. [25]; Ahmadireshnizi et al. [26]; Zhang et al. [27]	NL→model solver	General OR models	Model formulation; not a fixed policy family
<i>Data-driven inventory control without LLMs</i>			
Gabellini et al. [46]	Trained data-driven	Replenishment, safety stock and safety lead time	Closest OR neighbour; needs training; automotive data
Kolyaei et al. [35]; Stranieri et al. [36]; Wang et al. [37]	Deep reinforcement learning	Replenishment	Learned policies; require training environments
Ma [38]; Wang et al. [39]; Zhang et al. [40]	Deep learning / contextual DRO / online algorithm	Forecast-plus-order; capacitated replenishment	Training, tuning or worst-case design per setting
<i>Intermittent and spare-parts demand</i>			
Sanguri et al. [52]; Li et al. [53]; Goldsmith and Sachs [54]	Statistical / ML forecasting	Intermittent demand	Specialized estimators the LLM is not allowed to use
Babaveisi et al. [55]; Caserta and D’Angelo [56]	Optimization	Spare-part stocking	Multi-echelon or repairable settings; outside our single-echelon scope
Prak et al. [50]; Babai et al. [51]	Estimation / review	Inventory under intermittency	Regime-aware benchmarks for future work
InventOR (this paper)	(this Prompted LLM + deterministic gate and simulation)	(r, Q) per plant-material pair	No training; reviewed; information-controlled evaluation

215 4. The InventOR Decision Support System

216 4.1. Architecture and components

217 Table 4 maps the classical DSS components [10, 11] to their realization in
 218 InventOR and to the design requirements of Section 2.4. The generator is the
 219 only component whose internal logic is not inspectable, and it is surrounded
 220 by components that are deterministic and auditable. In the evaluated proto-
 221 type the dialog component is the per-pair output record and the comparison
 222 tables; an interactive planner interface was not built or evaluated.

Table 4: DSS components of InventOR.

Component	Realization in InventOR	Requirement
Data	Material-scoped extract of daily demand, lead times, working-day flags, cost and constraint parameters, and ERP reference values (Table 5); the set of fields supplied is controlled and logged	DR4
Knowledge	System prompt file defining the task, the prohibited methods, the storage rules and the output schema (Supplementary File S1)	DR1, DR5
Model: generator	Enterprise LLM application with a code interpreter, configured only by the prompt; proposes one (r, Q) policy, a justification and an escalation flag	DR1
Model: verifier and simulator	Deterministic parser and gate; held-out simulation with hard storage-limit replay; reference arms	DR2, DR3
Dialog (review)	Policy, justification, flag and simulated comparison presented to the planner under the trust, escalate and reject rules	DR6
Governance	Frozen prompt file, per-call log of input hash, time, session and raw response, repeated runs	DR5

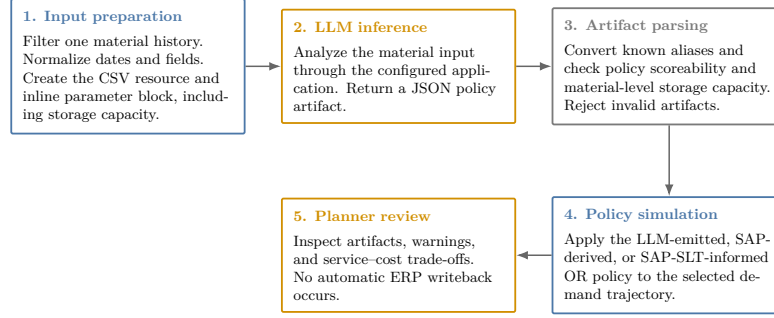
223 4.2. Operation in four steps

224 A planner’s question is: *given what this material has done in the past,*
 225 *how much safety stock should I hold, at what stock level should I reorder,*
 226 *and how much should I order?* InventOR answers this question in four steps
 227 (Figure 1).

228 **Step 1. Generate.** The LLM, accessed through an enterprise application
 229 with a code interpreter, receives one material’s history (daily de-

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

mand, realized lead times, working-day flags) and a short parameter block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [60] instructs it to compute demand and lead-time statistics, classify the demand pattern, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy together with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not

246 below the MOQ.

247 **Step 3. Simulate.** Each accepted policy is replayed on the material’s de-
248 mand observed *after* the data cutoff, a held-out evaluation in the
249 sense of out-of-sample forecast testing [61, 62], together with the
250 comparator policies, producing holding and ordering cost, fill rate
251 and stockout days. This plays the role that model-based simulation
252 plays in digital-twin decision support [22], at the scale of a single
253 policy.

254 **Step 4. Review.** A planner sees the policy, the LLM’s justification and
255 flag, and the simulated comparison. Only the planner changes ERP
256 parameters.

257 The role of the LLM is deliberately narrow. It is a *candidate generator*
258 whose output is never trusted without the gate, and it is not used to fore-
259 cast, to place orders, or to run the simulation. The prompt prohibits Monte
260 Carlo simulation, distribution fitting, machine-learning libraries and special-
261 ized intermittent-demand estimators, so that the policy rests on transparent
262 statistics a planner can reproduce.

263 4.3. Trust, review and reject rules

264 We propose three explicit rules for planners, in line with guidelines for
265 human–AI interaction and with internal-auditing practice for AI systems
266 [14, 15], and motivated by the evidence in Section 6.

267 **Trust (apply after review)** when the proposal passes the gate, the sim-
268 ulated trajectory respects the storage limit when orders are truncated at it,

269 and the simulated cost is not above that of the best history-only OR ref-
270 erence (Croston/SBA or empirical quantile) with a fill rate not more than
271 about 0.5 points lower.

272 **Escalate** when the LLM raises an escalation flag, the material is lumpy or
273 has strongly seasonal or trending demand, the plant or calendar data were
274 repaired, or the policy differs strongly between repeated generations.

275 **Reject** when the proposal fails the gate after the retry budget, exceeds the
276 storage limit in the hard-limit replay, or is dominated (higher cost and lower
277 fill) by a history-only reference.

278 4.4. *Provenance and reproducibility of the configuration*

279 The system prompt, which fully defines the task, the forbidden methods,
280 the capacity rules and the output schema, is a plain-text file that is deployed
281 unchanged in the enterprise application; it is distributed with the paper
282 (Supplementary File S1) and the code repository. Every call logs the SHA-
283 256 of its input data, the UTC time, the application session identifier and
284 the full raw response. The platform does not expose the underlying model
285 identifier or sampling settings to the client; the evaluated model (Anthropic
286 Claude Sonnet 4.6, with a code interpreter) is therefore declared by the
287 operator and is not machine-verifiable. We report this limitation openly and
288 base the reproducibility argument on the prompt file, the per-call logs and
289 three independent runs (Section 6.7) rather than on a model checkpoint.

290 5. Application and Evaluation Design

291 5.1. Evaluation approach

292 The evaluation addresses each design requirement in turn (Table 2):
293 whether the pipeline returns valid, feasible policies (DR1, DR2); what cost
294 and service the policies achieve against reference arms on held-out demand
295 (DR3); how much of that result is due to the information given to the LLM
296 rather than to its reasoning (DR4); whether storage limits hold along the
297 simulated trajectory (DR2); how stable policies are across independent runs
298 (DR5); and where proposals should be escalated (DR6). The primary basis
299 is the hard-limit replay with working-day receipts and no shortage penalty;
300 sensitivity analyses vary these choices. The evaluation is ex post and uses
301 no live users (Section 2.5).

302 5.2. Industrial application and data

303 The application is a pilot at a large manufacturer. The dataset contains
304 233,078 daily records for 411 plant-material pairs from three plants (A, B
305 and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon
306 observed to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring
307 (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For
308 19 pairs (all in Plant A) the supplied MOQ plus the reference safety stock
309 exceeds the storage limit, so no comparator can satisfy both rules; these are
310 excluded from outcome comparison, leaving a *common-feasible cohort* of 346
311 pairs (215, 65 and 66). All 365 pairs receive an LLM policy. Demand is
312 concentrated: the five largest materials account for 40.3% of demand and
313 the ten largest for 57.8%. The source working-day flag of Plant C is all zero;
314 we reconstruct it from the same-date majority of Plants A and B (ties count

as working days) and test the effect of this repair in Section 6.8. Such data-quality problems are a known threat to analytics-based decision support [63] and, for inventory systems, to record accuracy [64, 65, 66].

Table 5: Fields of the material-scoped CSV given to the LLM.

Column	Type	Description and use
date	date	Observation date; time ordering and aggregation
actual_demand	numeric	Realized daily demand; mean, variability, intermittency, regime
delivery_time	numeric	Recorded replenishment lead time (receipt rows)
is_working_day	binary	Working-day flag; demand statistics over working days
minimum_order_quantity	numeric	MOQ; order-size constraint
unit_cost,	numeric	Unit value, annual carrying-cost rate, fixed ordering cost;
holding_cost_rate,		EOQ and cost evaluation
ordering_cost		
service_level_target	numeric	Target service used in safety-stock computation
max_storage_units	numeric	Storage limit: $MOQ \leq Q$ and $SS + Q \leq \text{max_storage_units}$
safety_stock_units,	numeric	ERP planning parameters (current safety stock, safety lead time); withheld in the ablation arm
safety_time_workdays_actual		
reliability_calc,	mixed	Supplier and lane context
supplier_plant_distance,		
transportation_network		

5.3. Reference arms and information content

The LLM and the classical comparators do not necessarily use the same information, and this matters for the interpretation of any cost difference, because ERP safety stock and safety lead time embed accumulated planner judgement, and safety stock and safety lead time are alternative ways of buffering unreliability [45]. Planned lead times can also drift upward when they are adjusted dynamically (the “lead time syndrome” [67]), which would inflate any comparator that adds them to the lead time. We therefore construct arms that make the information explicit (Table 6).

Three further arms use no ERP input and represent stronger data-driven practice than the history-only OR. The *Croston/SBA OR* replaces the mean daily demand by the Croston estimate with the Syntetos–Boylan bias correction [47, 48] (smoothing constant 0.1, working-day demand history), uses

Table 6: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP used	inputs	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock		Status quo of the plant
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time		Original OR comparator
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both		OR arm matched to the LLM’s full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None		ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None		Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None		Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None		Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all fields of Table 5	Both		Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to “not supplied”	None		Ablation: what the LLM does when it cannot anchor on ERP

331 the root mean squared one-step forecast error as demand spread and the
 332 history-only lead-time moments, which suits intermittent demand [49]. The
 333 *empirical-quantile OR* sets the reorder point to the service-level quantile
 334 of the historical lead-time demand, computed from rolling sums over the
 335 rounded mean lead time in working days, and needs no distributional as-
 336 sumption. The *simulation-tuned (r, Q)* policy is a learned baseline: from a
 337 grid of reserve factors and order-quantity multiples around the history-only
 338 policy it selects the cheapest candidate whose fill rate on the last 250 work-

ing days *before the cutoff* meets the service target (or the best-fill candidate if none does). No post-cutoff data are used in any of these arms. We do not implement a deep reinforcement-learning baseline [36, 35]; the simulation-tuned policy is the trained comparator, and Section 8 lists neural baselines as a limitation.

5.4. Simulation, capacity replay and metrics

Each pair is simulated independently on the post-cutoff horizon as a single-item, single-echelon continuous-review (r, Q) system [57, 68, 69, 59, 58] with lost sales [70], in the spirit of simulation-based experiments on supply and timing uncertainty [71], a shared opening stock, an empty pipeline and arrivals after the rounded mean lead time in working days (Appendix Appendix A). MOQs are treated as hard order-size constraints, as imposed by suppliers [72], and storage limits as capacity bounds on the replenishment decision [40]. The primary metrics are the mean material-horizon holding-plus-ordering cost, the mean material-level fill rate, the aggregate (volume-weighted) fill rate and the number of stockout days. We report paired differences with 10,000-sample bootstrap 95% intervals.

A static check $SS + Q \leq \text{max_storage_units}$ does not show that storage limits are respected as the inventory evolves, because the reorder point includes lead-time demand and inventory on order can exceed the limit when it arrives. We therefore record, for every policy and every simulated day, the post-receipt on-hand stock and report the share of pairs and days above the limit (the *unconstrained* simulation). We then *replay* every policy with each order truncated to the storage limit minus the current inventory position (the *hard-limit replay*). The hard-limit replay is our primary evaluation.

364 5.5. *Repeated runs and sensitivities*

365 To examine run-to-run variation, the entire cohort is generated three
366 times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1
367 serves as the reference for tabulated results unless stated otherwise; runs 2
368 and 3 provide replicates. Sensitivity analyses vary the working-day recon-
369 struction (majority, any-plant, strict-all, exclusion of Plant C and calendar-
370 day arrivals), the shortage penalty (zero, one and five times unit cost), and
371 the exclusion of diagnostic outliers (52 pairs with coefficient-of-variation er-
372 ror above 0.75 or safety stock outside half to twice the reference).

373 6. Results

374 6.1. *Completion and policy characteristics*

375 All four runs (three with the full input and one ablation, Section 6.4)
376 return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most
377 three attempts per pair. Retries were mostly caused by schema errors (invalid
378 policy), not by capacity violations. The validity gate rejects non-positive
379 reorder points and order quantities, negative safety stocks and reorder points
380 below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs
381 (188 of the 346 in the cohort below), so for a majority of pairs the SAP-
382 derived arm holds no safety stock at all.

383 6.2. *Headline comparison with ERP-derived parameters*

384 Unless stated otherwise, all results from here on use the hard-limit replay
385 (every order truncated at the storage limit) on the 346-pair common-feasible
386 cohort, with working-day receipts and no shortage penalty; Section 6.5 re-
387 ports the unconstrained simulation for comparison. Table 7 compares all

arms. In run 1 the LLM policies cost 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed OR comparator (mean 2,928 versus 3,305 and 4,211). The mean fill rate is 0.18 points lower than for SAP-derived (98.11% versus 98.30%; bootstrap interval -0.54 to $+0.10$) and 0.11 points lower than for the SLT-informed OR. The mean cost difference against SAP-derived is -377 (95% interval -651 to -154); the LLM is cheaper than SAP-derived in 241 of 346 pairs and, in 226 pairs, simultaneously no more expensive and no worse in fill. The LLM incurs more stockout days in total (910 versus 804 and 831), and aggregate fill is 78.87% versus 79.34% and 79.15%. Aggregate fill is dominated by a few high-demand materials: in the unconstrained simulation of run 1, excluding the five largest materials raises all three to 94.6–95.8% (Figure 2).

The cost decomposition shows where the saving comes from. Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and has about 16% lower holding cost, while it places about the same number of orders per year (17.3 versus 17.6). The saving is therefore a trade of inventory for service: the LLM holds less stock and accepts about 13% more stockout days. It is not the consequence of a better ordering rhythm, and it is not free of service cost. Section 6.8 shows that, once shortages carry a cost, this trade no longer favours the LLM.

Table 7: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year.

Arm	Mean cost	Holding	Ordering	Avg. inventory	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days
SAP-derived	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890

Demand concentration and fill sensitivity

Largest-demand materials strongly influence full-cohort aggregate fill.

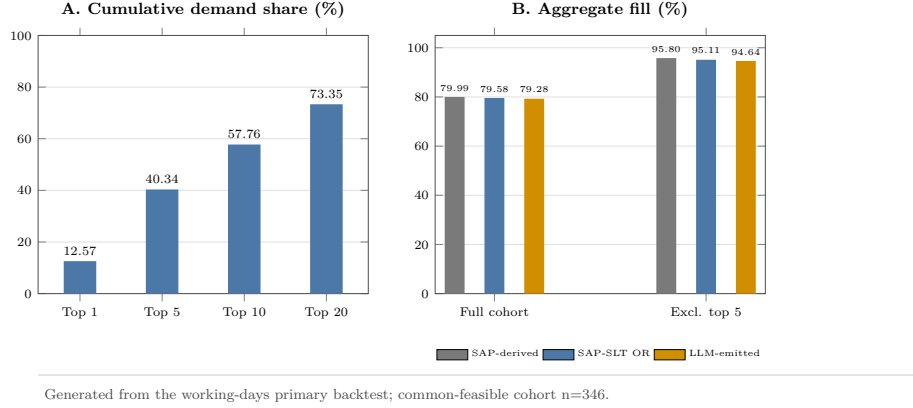


Figure 2: Demand concentration and aggregate-fill sensitivity (unconstrained simulation, run 1). Panel A: cumulative demand share of the largest materials. Panel B: aggregate fill for the full cohort and after excluding the five largest materials.

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

Table 8 compares the arms in the hard-limit replay on the 346 pairs of the cohort, for all three runs. Three results stand out.

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248). The ERP safety lead time inflates their effective lead time, hence their buffers; against them the LLM is about 30–31% cheaper in every run. This comparison therefore mainly measures how heavily ERP inputs over-buffer, not the quality of the LLM.

Second, the *history-only OR*, which uses no ERP inputs, has a cost of 2,764 at a fill of 98.14%. Against it the LLM is 5.9–6.0% more expensive in all three runs (run 1: mean difference +164, 95% interval +91 to +240), with fill differences of -0.03 , -0.25 and $+0.01$ points whose intervals include zero in all three runs. The LLM is cheaper than the history-only OR in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap. The Croston/SBA OR (2,705, fill 98.30%) is 8.2–8.4% cheaper than the LLM (run 1: mean

423 difference +223, interval +97 to +362; the LLM is cheaper in 91–99 pairs),
 424 and the empirical-quantile OR (2,489, fill 98.35%) is about 15% cheaper, that
 425 is, the LLM costs 17.7–17.8% more (mean difference +439, interval +329 to
 426 +557; the LLM is cheaper in 44–49 pairs) with a mean fill that is 0.24 points
 427 lower. Part of the LLM’s lower cost relative to ERP therefore reflects the
 428 weakness of the ERP benchmarks and not LLM superiority; the LLM does
 429 not match a competent data-driven (r, Q) baseline.

430 The simulation-tuned (r, Q) policy sits at a different operating point. It
 431 reaches an aggregate fill of 84.8% with the lowest average inventory (11,914
 432 units) but a lower mean fill (96.05%) and many more, smaller stockouts
 433 (1,973 stockout days), because its tuning criterion is the fill of the last pre-
 434 cutoff year, not the post-cutoff horizon. Against it the LLM is 10.3–10.4%
 435 cheaper in the zero-penalty cost, with a 2.1-point higher mean fill, but the
 436 paired interval of the cost difference includes zero (−2,527 to +857 in run 1).
 437 Once shortages are priced it becomes by far the cheapest arm (Section 6.8),
 438 so its low zero-penalty ranking is not a sign of a weak baseline.

439 Third, the LLM is nonetheless cheaper than ERP-derived parameters
 440 (11.3–11.4%) at a mean fill that is lower by 0.14–0.40 points (the interval
 441 excludes zero only in run 2), so it is useful when the alternative is the status
 442 quo. It does not approach the best classical baselines without code or train-
 443 ing, but it produces a reviewable policy for every material without either.
 444 Whether the LLM relies on the ERP planning parameters it is given is tested
 445 in the ablation of Section 6.4.

446 6.4. LLM without ERP planning parameters (ablation)

447 In this ablation the LLM receives exactly the same material histories,
 448 cost parameters, MOQ, service target and storage limit as in runs 1–3, but

Table 8: Hard-limit replay: mean cost and mean fill by arm, $n = 346$; LLM cost relative to each arm (negative = LLM cheaper).

Arm	Mean cost	Mean fill (%)	LLM cost vs arm (%)		
			Run 1	Run 2	Run 3
SAP-derived	3,305	98.30	−11.4	−11.3	−11.4
SLT-informed OR	4,211	98.23	−30.5	−30.4	−30.5
ERP-floor OR	4,248	98.23	−31.1	−31.0	−31.1
History-only OR	2,764	98.14	+5.9	+6.0	+5.9
Croston/SBA OR	2,705	98.30	+8.2	+8.4	+8.3
Empirical-quantile OR	2,489	98.35	+17.7	+17.8	+17.7
Simulation-tuned (r, Q)	3,266	96.05	−10.4 [‡]	−10.3 [‡]	−10.3 [‡]
LLM (runs 1, 2, 3)	2,928 / 2,931 / 2,929	98.11 / 97.89 / 98.16			
LLM without ERP (ablation)	3,014	98.06	−8.8 [†]		

[†] Relative to SAP-derived; other comparisons are in Section 6.4. [‡] Different operating point (mean fill 96.05%, aggregate fill 84.8%); the cost difference has a paired interval that includes zero, and with priced shortages the policy is the cheapest arm (Section 6.8).

the ERP safety stock and ERP safety lead time are blanked in the uploaded CSV and marked “not supplied” in the inline parameter block; the system prompt file is unchanged. It tests whether the LLM anchors on the ERP planning parameters it is given, and therefore whether the comparison with the ERP-informed arms is confounded by information. The ablation run has the same structure as runs 1–3: all 365 pairs returned a valid, storage-compliant policy.

Policy values. Withholding the ERP parameters changes the LLM’s policies only slightly beyond its ordinary run-to-run variation. The median ratios of safety stock and reorder point to run 1 are both 1.00 (346 common pairs). The reorder point differs by more than 10% in 54.6% of pairs, against 48.3% between runs 1 and 3 and 48.0% between runs 2 and 3 with the ERP parameters supplied; the safety stock differs by more than 10% in 11.0% of pairs, against 9.8% and 9.5%. The median absolute reorder-point difference is 128 units, against 90 and 84 between ERP-supplied runs. The mean ratio is slightly above one (1.04 for safety stock, 1.05 for the reorder point), so the LLM tends to buffer a little more when it has no ERP anchor.

466 *Outcomes.* In the hard-limit replay the LLM without ERP parameters
467 has a mean cost of 3,014 at a mean fill of 98.06%, against 2,928–2,931 and
468 97.89–98.16% in runs 1–3. It remains 8.8% cheaper than SAP-derived pa-
469 rameters (95% interval of the mean difference -566 to -61 ; fill -0.24 points,
470 interval including zero) and 28.4–29.1% cheaper than the SLT-informed and
471 ERP-floor ORs, but it is 9.0% more expensive than the history-only OR
472 (interval $+134$ to $+387$; fill -0.09 points; cheaper in 91 of 346 pairs), 11.4%
473 more expensive than the Croston/SBA OR and 21.1% more expensive than
474 the empirical-quantile OR, and 7.7% cheaper than the simulation-tuned pol-
475 icy at its different operating point. In the unconstrained simulation the cor-
476 responding figures are -7.0% against SAP-derived (interval -513 to $+16$,
477 which includes zero), -23.2% and -23.8% against the two ERP-informed
478 ORs, and $+8.8\%$ against the history-only OR. Withholding the ERP pa-
479 rameters thus raises the LLM’s cost by roughly 3 percentage points relative
480 to the history-only OR and does not reverse any ranking.

481 *By plant.* The pattern is the same as with ERP inputs: the LLM remains
482 cheaper than SAP-derived at Plants A (215 pairs, -10.8%) and B (65 pairs,
483 -21.9%) and more expensive at Plant C (66 pairs, $+41.6\%$ against $+36.2\%$
484 in run 1; cheaper in 8 pairs). The Plant C loss is therefore not caused by
485 the LLM leaning on ERP parameters either.

486 *Interpretation.* Two conclusions follow. First, the information-
487 asymmetry concern (that the LLM benefits from ERP inputs) does not
488 explain the LLM’s performance: with the ERP parameters removed the
489 LLM keeps most of its advantage over ERP-derived parameters, and the
490 gap to the history-only OR widens only moderately. Second, the ERP
491 parameters are of limited use to the LLM; most of its output is determined
492 by the demand and lead-time histories. A practical consequence is that the

LLM can be deployed for materials whose ERP planning data are missing or stale. One limitation is that the ablation tests only the two ERP planning fields; the LLM still sees unit cost, holding rate, ordering cost, MOQ, service target and storage limit, which all reference arms also use.

6.5. Storage limits along simulated trajectories

All 365 LLM policies satisfy the static rule $SS+Q \leq \text{max_storage_units}$. Along the simulated trajectories, however, the unconstrained simulation exceeds the limit on post-receipt on-hand stock for 183 pairs (52.9%) for the LLM (10.2% of simulated pair-days; median peak 1.02 times the limit; maximum 2.52), 198 pairs (57.2%; 15.9% of days) for SAP-derived and 230 pairs (66.5%; 23.4% of days) for the SLT-informed OR; no pair starts above the limit. The three LLM runs give 183, 181 and 178 pairs. Exceedances arise because inventory on order and lead-time demand can add to stock above $SS+Q$; the static check is therefore necessary but not sufficient, and it applies to every method, not just the LLM.

Under the hard-limit replay no arm exceeds the limit by more than one unit (rounding). Enforcing the limit raises the LLM mean cost from 2,850 to 2,928 and lowers its mean fill by 0.04 points (98.15% to 98.11%); SAP-derived rises from 3,184 to 3,305 and the SLT-informed OR from 3,852 to 4,211, with the latter placing 24.5 instead of 15.8 orders on average. The ranking among arms is unchanged, and the LLM’s relative advantage over SAP-derived is 11.4% instead of 10.5%. We therefore use the replay as the primary evaluation.

6.6. Plants and demand regimes

The average hides a strong plant effect (Table 9, hard-limit replay, run 1). The LLM is 14.1% cheaper than SAP-derived in Plant A (cheaper in 179 of

519 215 pairs) and 22.7% cheaper in Plant B (54 of 65), but 36.2% *more* expensive
 520 in Plant C, where it is cheaper in only 8 of 66 pairs. The Plant C loss is
 521 similar in the other full runs (+32.2% and +31.5% in runs 2 and 3) and
 522 larger without the ERP parameters (+41.6%). Plant C is also the plant
 523 whose calendar we repaired; Section 6.8 shows that the loss persists under
 524 every working-day reconstruction in which Plant C can be scored. Against
 525 the history-only OR the LLM is 7.2% costlier in Plant A, 3.6% in Plant B and
 526 6.7% in Plant C (run 1), against the Croston/SBA OR 13.1%, 2.3% and 6.2%
 527 costlier (cheaper in 26 of 65 Plant B pairs and 20 of 66 Plant C pairs), and
 528 against the empirical-quantile OR 20.8%, 7.7% and 29.5% costlier. No plant
 529 therefore favours the LLM over a data-driven (r, Q) baseline, and Plant B is
 530 the plant where it comes closest. Note that the SLT-informed and history-
 531 only ORs give identical results in Plant C, because the ERP safety lead time
 532 is zero for all 66 Plant C pairs.

533 By demand regime, the 33 lumpy materials show the largest cost re-
 534 duction against SAP-derived (-25.4% in run 1; -23.7% to -25.2% in the
 535 other runs), whereas the 313 smooth materials show -9.8% at a 0.24-point
 536 lower mean fill (98.64% versus 98.88%). The service effect for lumpy ma-
 537 terials is not consistent (mean fill $+0.38$ points in run 1, -0.29 in run 2),
 538 and the history-only OR is again 9% cheaper than the LLM there while
 539 achieving a much lower aggregate fill (54.3% versus 57.0% for the LLM and
 540 65.8% for SAP-derived), so part of any cost saving on lumpy materials is
 541 bought with service. The Croston/SBA OR is 9.7% cheaper than the LLM
 542 on these materials with a higher mean fill (94.55% versus 93.12%), whereas
 543 the empirical-quantile OR is 1.2% more expensive (the LLM is cheaper in 15
 544 of 33 pairs); the LLM therefore matches the best data-driven baseline only
 545 on lumpy materials, and not the intermittent-demand baseline designed for

546 them. With only 33 lumpy materials we regard this as indicative only. By
547 demand variability (terciles of the coefficient of variation), the LLM is 15.7%
548 cheaper than SAP-derived at equal fill in the lowest tercile, 7.5% cheaper with
549 a 0.68-point lower mean fill in the middle tercile (98.10% versus 98.78%), and
550 12.9% cheaper in the highest tercile.

Table 9: Mean simulated cost per pair and mean fill by plant, run 1, hard-limit replay (common-feasible cohort). The SLT-informed and history-only ORs coincide in Plant C.

Plant (n)	LLM cost	SAP-derived cost	SLT-informed OR cost	History-only OR cost	Croston/SBA OR cost	Empirical quantile OR cost	LLM vs SAP	Cheaper pairs	Fill LLM / SAP (%)
All (348)	2,928	3,385	4,211	2,764	2,705	2,889	-11.4%	241	98.11 / 98.20
A (215)	2,391	2,782	3,770	2,231	2,114	1,980	-14.1%	170	98.18 / 98.57
B (65)	4,916	6,360	7,329	4,746	4,866	4,566	-22.7%	54	98.23 / 96.47
C (68)	2,719	1,897	2,549	2,549	2,561	2,100	+36.2%	8	96.50 / 96.61

551 6.7. Run-to-run stability

552 Across the three runs the aggregate results are very similar: LLM cost
553 relative to SAP-derived is -10.5%, -9.8% and -10.2% in the unconstrained
554 simulation, and -11.4%, -11.3% and -11.4% in the hard-limit replay, so
555 the spread is below one percentage point. The distributions of policy values
556 are near-identical (median reorder point 1,584, 1,582 and 1,466; median or-
557 der quantity 2,737, 2,777 and 2,693; median safety stock 683, 688 and 669;
558 Figure 3). Individual policies, however, vary. Comparing runs 2 and 3 on
559 the 365 pairs, all three policy values are identical for only 36 pairs; every
560 policy resolves to the (r, Q) family, and the median absolute differences are
561 81 units for the reorder point, 6 for the order quantity and 3 for the safety
562 stock. A few pairs vary strongly (for one material the reorder point was
563 14,882, 14,947 and 29,218 units in the three runs; for two others the safety
564 stock was zero in one run and 660 or 1,075 units in the others). The practical
565 implication is that one generation per material is a noisy estimate whose ag-
566 gregate behaviour is stable but whose individual values should be reviewed.
567 Non-determinism of this kind is a documented property of LLMs [16], so
568 that repeated-run reporting should be standard.

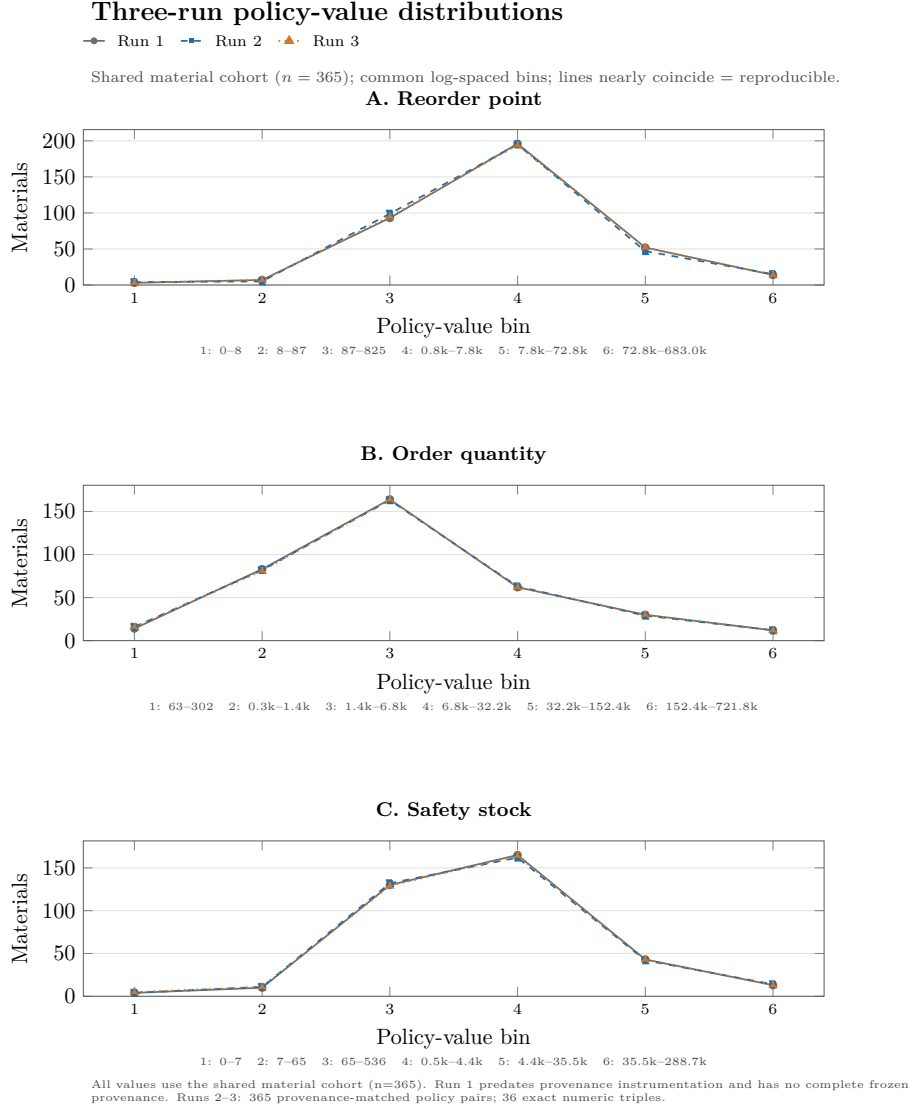


Figure 3: Policy-value distributions across the 365 shared pairs for runs 1, 2 and 3 (materials per common log-spaced bin); near-coincident lines indicate that aggregate behaviour is reproducible.

569 *6.8. Sensitivity analyses*

570 *Plant C calendar..* Plant C’s source working-day flag is all zero, which re-
571 quired reconstruction. We re-scored all policies (hard-limit replay) under

572 four alternatives: any-plant, strict-all, exclusion of Plant C and calendar-
 573 day arrivals. In every variant in which Plant C can be scored (any-plant and
 574 calendar-day arrivals), the LLM remains 31–35% more expensive than SAP-
 575 derived in Plant C across the three full runs. The cohort-wide advantage
 576 over SAP-derived is -10.8% to -11.1% in these variants (primary calen-
 577 dar: -11.3% to -11.4%), and the cost excess over the history-only OR is
 578 $5.6\text{--}6.1\%$. The strict and exclusion variants (280 pairs) drop Plant C and
 579 show a larger LLM advantage over SAP-derived (-17.0% to -17.6%) and
 580 a $5.8\text{--}6.6\%$ excess over the history-only OR. The Plant C result is therefore
 581 not an artefact of our calendar repair. One residual risk remains: the LLM
 582 saw demand series for Plant C that were themselves built on the repaired
 583 calendar, and only a re-generation with alternative inputs could remove that
 584 dependence.

585 *Shortage penalty.* With a positive lost-sales penalty the zero-penalty cost
 586 ranking changes. Table 10 gives mean total cost per pair (holding, ordering
 587 and shortage cost) in the hard-limit replay for penalties of zero, one and
 588 five times the unit cost. The simulation-tuned policy is by far the cheapest
 589 arm as soon as shortages are priced (1,094,043 at one times the unit cost,
 590 about 33% below SAP-derived), because it was selected to meet the service
 591 target on the pre-cutoff history. Among the other arms the SAP-derived
 592 parameters are the cheapest, because they carry more inventory and fewer
 593 stockouts. Run 1 of the LLM then costs 2.4% more than SAP-derived at both
 594 one and five times the unit cost; it is about as expensive as the Croston/SBA
 595 OR (1,673,003 versus 1,674,925 at one times the unit cost), and it remains
 596 cheaper than the SLT-informed OR (1.2% lower at both multiples), the ERP-
 597 floor OR, the empirical-quantile OR and slightly cheaper than the history-

only OR (0.8% lower at both multiples). The LLM’s cost advantage over ERP-derived parameters is therefore specific to the case in which service shortfalls are not charged, and the service deficit is a real cost when shortages are priced.

Table 10: Mean total cost per pair in the hard-limit replay (346 pairs) when lost sales are charged at a multiple of the unit cost.

Arm	0×	1×	5×
SAP-derived	3,305	1,634,136	8,157,462
SLT-informed OR	4,211	1,694,183	8,454,072
ERP-floor OR	4,248	1,694,220	8,454,109
History-only OR	2,764	1,687,191	8,424,899
Croston/SBA OR	2,705	1,674,925	8,363,806
Empirical-quantile OR	2,489	1,693,784	8,458,966
Simulation-tuned (r, Q)	3,266	1,094,043	5,457,151
LLM run 1	2,928	1,673,003	8,353,306
LLM run 2	2,931	1,683,280	8,404,676
LLM run 3	2,929	1,671,759	8,347,082
LLM no ERP	3,014	1,689,961	8,437,748

Diagnostic outliers and calendar arrivals.. In the unconstrained simulation of run 1, removing the 52 diagnostic outliers (314 pairs; coefficient of variation error above 0.75 or safety stock outside half to twice the reference) leaves LLM, SAP-derived and SLT-informed OR aggregate fill at 79.96%, 80.38% and 79.99%, with mean cost 2,864, 3,098 and 3,748, so the ordering of the arms is unchanged. The findings thus do not depend on a few extreme cases.

7. Managerial Insights

1. Position the tool as a reviewed decision support system, not an autonomous policy engine. Single generations differ at the material level, and roughly 6–8% of outputs carry a free-text escalation flag. The gate-simulate-review structure (Section 4.3) captures most of the value of the LLM (a candidate policy for every material, with a justification) while leaving the decision to the planner.

- 616 **2. Benchmark against data-driven OR baselines before claiming**
617 **value.** ERP-derived parameters and ERP-informed OR policies are weak
618 benchmarks: safety stocks are often stale or zero and safety lead times in-
619 flate the effective lead time. A planning team that evaluates an LLM aid
620 only against ERP will overestimate its benefit. History-only (r, Q) base-
621 lines (normal approximation, Croston/SBA, empirical lead-time-demand
622 quantile) are cheap to compute and were cheaper than the LLM in our
623 data, by 6%, 8% and 18% respectively; a policy tuned by simulation on
624 the pre-cutoff history was cheapest once shortages were priced.
- 625 **3. Where the LLM is useful.** Plants A and B (cost reductions of 14-23%
626 against ERP), lumpy materials (about 25% cheaper than ERP-derived
627 parameters, although with mixed service effects and only 33 materials)
628 and the many materials with no analyst time are the sweet spot. Plant C,
629 where the LLM loses, indicates that conditions with repaired calendars,
630 small samples or atypical lead-time behaviour need escalation.
- 631 **4. Verify storage along the trajectory.** The static rule $SS + Q \leq \text{limit}$
632 cannot protect a warehouse: 53-66% of pairs breach the limit in simula-
633 tion under every method. Operations should enforce a cap in the order-
634 release logic (order-up-to or truncation), or treat the limit as a design
635 requirement in the policy.
- 636 **5. Price shortages explicitly.** Without a shortage penalty the cost rank-
637 ing rewards under-stocking. With a penalty of one unit cost the LLM's
638 cost advantage over SAP disappears (it is about 2.4% more expensive).
639 Planners should set service targets and penalties before reading the cost
640 comparison.

641 **6. Freeze and log the configuration.** Treat the system prompt as a
642 controlled file, log every call (input hash, time, session, raw response)
643 and re-run before changing the prompt. Our reproducibility argument
644 rests on this log, not on the platform’s model identifier, which the client
645 cannot see.

646 **7. Cost and latency.** Each LLM generation took about 7–20 minutes,
647 whereas the OR baseline takes milliseconds. The LLM is attractive where
648 analyst time and not compute is scarce, but a deployment needs batching
649 and monitoring. Periodic regeneration as demand and lead times change
650 would also support adaptable planning under uncertainty [73].

651 **8. Discussion, Limitations and Research Perspectives**

652 *What the evidence shows..* In a three-plant industrial dataset, a prompt-
653 configured LLM generates feasible, reviewable (r, Q) policies for every ma-
654 terial. They are cheaper than ERP-derived parameters in the simulation,
655 but they do not beat history-only data-driven OR policies (Croston/SBA,
656 empirical-quantile). We therefore describe InventOR as a way to make a
657 competent analyst’s workflow available at scale, not as a method that out-
658 performs optimization. The information-controlled evaluation is what al-
659 lows this conclusion; without the ERP-independent arms, the first-cut result
660 (−11.4% against SAP and −30.5% against the SLT-informed OR in the
661 hard-limit replay) would have been presented as superiority.

662 *Design principles for LLM-based decision support..* The evidence supports
663 six principles that go beyond this application and that we offer as a basis
664 for design theory [7]. **DP1, bound the generator:** give the LLM the

665 role of candidate generator with a prompt that fixes the method, the con-
 666 straints and the output schema, and no authority to act. **DP2, verify**
 667 **deterministically and along time:** check every output with code, and
 668 check feasibility along the trajectory, because point checks of constraints
 669 can pass while the constraints are violated over time (Section 6.5). **DP3,**
 670 **show counterfactual evidence:** give the reviewer a simulated compari-
 671 son with credible references, so that review is based on outcomes and not
 672 on the persuasiveness of the justification. **DP4, control information in**
 673 **evaluation:** compare against baselines that use the same information and
 674 against a baseline that uses none, and withhold incumbent inputs from the
 675 LLM in an ablation, otherwise a gain may merely reflect a weak benchmark
 676 (Sections 6.3, 6.4). **DP5, freeze and log:** treat the prompt as a controlled
 677 file, log every call and repeat the runs, because individual outputs vary (Sec-
 678 tion 6.7). **DP6, escalate by rule:** define in advance when proposals are
 679 trusted, escalated or rejected, to limit complacency [13] (Section 4.3).

680 *Limitations..* (i) The data come from one manufacturer and three plants;
 681 generalization to other settings is not shown. (ii) The platform does not
 682 expose the model identifier or inference settings, so the model is operator-
 683 declared. (iii) The simulation excludes observed receipts, corrections and
 684 scrap and starts with an empty pipeline; a prospective field trial with plan-
 685 ners is needed to assess realized benefit, which we do not claim. (iv) The
 686 cost model has zero shortage penalty in the primary results and standard
 687 holding and ordering costs; with a positive penalty the cost ranking changes.
 688 (v) The (r, Q) policy is a single-item, single-echelon family; multi-item cou-
 689 pling and shared capacity are outside the scope. (vi) Run-to-run variation of
 690 individual policies is substantial. (vii) Plant C required calendar reconstruc-

tion. (viii) No planner outcomes were measured and the dialog component is a prototype record, not an evaluated interface: we make no claim about usability, adoption, planner workload or decision quality, which a DSS evaluation in the field would need to establish. (ix) The reference arms comprise ERP-derived and normal-approximation (r, Q) policies, Croston/SBA and empirical-quantile (r, Q) policies and a simulation-tuned (r, Q) policy; we did not include deep reinforcement-learning or neural forecasting baselines, and the simulation-tuned policy is a lightweight trained comparator, so the comparison with the state of the art is still incomplete. (x) No field study with planners was conducted; planner time, override behaviour and adoption are not measured.

Research perspectives..

1. *Ensembling and consistency.* Aggregating several generations (median or consensus policies) may reduce the individual-policy noise while retaining the LLM’s justification; the copilot findings for spare parts [34] suggest this is promising.
2. *Hybrid generators.* Combining the best data-driven OR policy (Croston/SBA or empirical quantile) with the LLM (the LLM reviewing and annotating the OR policy, or choosing among candidates by regime), as proposed for LLM–OR integration [74], could capture the OR’s cost level and the LLM’s flags.
3. *Comparison with trained methods.* A benchmark that compares prompt-based generators with deep reinforcement-learning and other trained policies [46, 35, 36] on identical data and information would clarify the trade-off between setup effort and policy quality.

716 4. *Intermittent-demand and richer policy families.* Allowing order-up-to or
 717 (s, S) policies and regime-specific estimators may help the lumpy and
 718 Plant C pairs [50, 51].

719 5. *Planner-in-the-loop field trials.* Prospective studies should compare plan-
 720 ners working with and without the aid on a matched set of materials, and
 721 measure planner time, override rates and realized service and inventory,
 722 which our simulation cannot, and should examine adoption drivers and
 723 barriers with mixed-methods designs [20, 19, 5].

724 6. *Standards for information-controlled evaluation.* Studies of LLM-based
 725 decision support systems should report an ERP-independent baseline, an
 726 ablation that withholds the incumbent’s inputs, trajectory-level feasibil-
 727 ity, and run-to-run variation.

728 9. Conclusion

729 We defined InventOR as a human-reviewed decision support system for a
 730 recurring production-planning decision, the setting of replenishment param-
 731 eters per plant-material pair, and evaluated it against six design requirements
 732 on a three-plant industrial dataset. A prompt-configured LLM generated
 733 feasible, justified (r, Q) policies for every material without training, and a
 734 deterministic gate, a hard-limit simulation and explicit review rules made
 735 these proposals checkable by a planner. The proposals cost about 11% less
 736 than ERP-derived parameters at slightly lower service, but about 6–8% more
 737 than a history-only or Croston/SBA OR policy and about 18% more than
 738 an empirical-quantile policy; when lost sales are priced at one unit cost or
 739 more, ERP-derived parameters are the cheapest of the non-tuned arms and
 740 a simulation-tuned policy is cheapest overall. Storage limits that hold in

741 a static check are exceeded along simulated trajectories for every method,
 742 and the LLM loses to ERP at one plant. The value of the system there-
 743 fore lies in its role as an auditable candidate generator inside a reviewed
 744 process, benchmarked against history-only OR, with trajectory-level storage
 745 checks, explicit shortage pricing and a frozen, logged configuration, and not
 746 in out-optimizing a competent data-driven OR baseline.

747 For the wider production-research community, the contribution is a set
 748 of design principles (Section 8) that apply to any planning decision in which
 749 parameters are set item by item and an LLM could be asked to propose
 750 them: bound the generator, verify deterministically and along time, show
 751 counterfactual evidence, control information in evaluation, freeze and log the
 752 configuration, and escalate by rule. These results are simulated-comparison
 753 evidence for an artifact; field evidence on planner behaviour and realized
 754 outcomes remains the central open step.

755 **Appendix A. Formal Definitions**

756 *Reference calculation..* These are the standard textbook calculations [59, 68].
 757 For pair i , let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff
 758 working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the
 759 lead time in working days, K_i the ordering cost, h_i the annual holding cost
 760 per unit (unit cost times holding rate), z_i the standard normal quantile of
 761 the service target, and 250 the working days per year. The order quantity
 762 is $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i \cdot 250\bar{d}_i/h_i} \rceil\}$. The lead-time-demand standard
 763 deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i} \rceil$
 764 and the reorder point $R_i = \lceil \bar{d}_iL_i^{\text{eff}} + SS_i \rceil$, where L_i^{eff} is the mean lead time
 765 for the history-only OR and the mean lead time plus the ERP safety lead

time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm uses the ERP safety stock and the same Q_i . All policies are projected to $SS_i + Q_i \leq \text{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

Simulation.. On each post-cutoff working day, demand is served from on-hand stock, unmet demand is lost, and an order of size Q_i is placed when the inventory position (on-hand plus on order) is at or below R_i and arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. The total cost is $TC_i = h_i \bar{I}_i T / 250 + K_i N_i$, where $\bar{I}_i$ is the mean on-hand stock over the T simulated working days and N_i the number of orders. Fill rate is served divided by demanded units. In the hard-limit replay an order is truncated to $\text{max_storage_units}_i$ minus the inventory position, and no order is placed if the truncated quantity is zero.

Appendix B. Reproducibility Record

Prompt file, runner code, evaluation scripts, aggregate outputs and per-call input hashes are distributed with the code release (Data and Code Availability). The three runs and the ablation use the same prompt file; the ablation differs only in that the two ERP planning fields are blanked in the uploaded data and marked “not supplied” in the inline block. The deployed system prompt is the file `system_prompt.txt` in the release. The platform does not expose the model identifier or inference settings to the client; the model is therefore operator-declared (Anthropic Claude Sonnet 4.6 on an enterprise platform with a code-interpreter tool) and not independently verified. For each call the release records the input data, a hash of the input, the UTC time, the session identifier and the raw response.

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799 The author declares no known competing financial interests or personal
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801 paper.

802 **Data and Code Availability**

803 The raw operational data and material-level artifacts cannot be publicly
804 released because they contain commercially sensitive material, supplier
805 and planning information. A confidentiality-screened release package
806 containing analysis code, the anonymized variable schema, aggregate
807 evaluation results, manuscript sources and prompt files is available at
808 github.com/DressPD/inventor_public. A versioned Zenodo archive of
809 the release is available at <https://doi.org/10.5281/zenodo.22262052>. A
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811 Supplementary File S1 contains the system prompt used for all runs.

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818 **Declaration of Generative AI and AI-Assisted Technologies in the** 819 **Writing Process**

820 During manuscript preparation and revision, the author used OpenAI
821 GPT-5.6 and Anthropic Claude models through the OpenCode coding assis-
822 tant. They supported language editing, restructuring and redrafting of the
823 text, consistency checks, citation verification against Crossref metadata, La-
824 TeX proofing, and the writing and execution of the analysis scripts that pro-
825 duce the evidence tables (for example the capacity-trajectory, information-
826 asymmetry and sensitivity analyses). These writing and coding assistants
827 are distinct from the Anthropic Claude Sonnet 4.6 deployment that is the
828 object of evaluation in the research workflow. The author specified the anal-
829 yses, reviewed and edited all content and results after use, and takes full
830 responsibility for the content of the published article. Figures were rendered
831 deterministically from author-reviewed LaTeX and Python source; genera-
832 tive AI did not directly create or manipulate image content.

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